

COMPSCI 752

Big Data Management

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Some Solutions to Strategic Exercises for XML

- ① Explain the difference in purpose between the HTML and XML languages.
- ② Briefly explain two main *application areas* of XML.
- ③ Name four reasons for the *success of XML*.

Difference between HTML and XML

HTML

- HTML is designed to display data and focus on how data looks.
- HTML is about presentation

Difference between HTML and XML

HTML

- HTML is designed to display data and focus on how data looks.
- HTML is about presentation

XML

- XML is designed to describe data and focus on what data is
- XML is about content

Application areas of XML

- XML can separate data from its presentation
- XML is the industry standard for the exchange of data on the Web and elsewhere
- XML can be used to store data and roll-out data into different format needs (transformation)
- XML can be used to create new application-specific languages (dialects), such as business models
- XML enables Web publishing
- XML forms a foundation for data integration
- XML forms a foundation for distributed computations, such as Web services

Reasons for success

- being there at the right time (Internet bubble)
- politically correct (the W3C is OS-agnostic)
- technically sound and simple
- human-readable based on a well-known syntax
- great for rapid prototyping and experiments

- 1 Name two basic rules that *well-formed XML documents* must satisfy.
- 2 For each rule, give a simple example that violates this rule.

Rules for XML documents to be *well-formed*

- start and end tags must match, e.g.

`<boat> </boot>` ~~✗~~

is not permitted

Rules for XML documents to be *well-formed*

- start and end tags must match, e.g.

`<boat> </boot>`

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- start and end tags must nest properly, e.g.

`<boat> <boot> </boat> </boot>`



X

is not permitted

Rules for XML documents to be *well-formed*

- start and end tags must match, e.g.

`<boat> </boot>`

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- start and end tags must nest properly, e.g.

`<boat> <boot> </boat> </boot>`

is not permitted

- XML documents have exactly one element root, e.g.

`<root1> </root1> <root2> </root2>` 

is not permitted

Structure of XML documents

Consider the following XML document.

```
<?xml version="1.0" encoding="ISO-8859-1" ?>
```

```
<?xml-stylesheet href="headlines.css" type="text/css" ?>
```

```
<invoice>  
  <product>Lenovo X201</product>  
  <customer>Jack Ryan</customer>  
  <amount>NZ$1822</amount>  
  <data>30-03-2011</date>  
</invoice>
```

Indicate the *major parts* of the XML document above.

Parts of XML documents

`<?xml version="1.0" encoding="ISO-8859-1" ?>`

declaration

`<?xml-stylesheet href="headlines.css" type="text/css" ?>`

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<invoice>
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Parts of XML documents

`<?xml version="1.0" encoding="ISO-8859-1" ?>`

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pro logue

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`</invoice>`

XML declaration

Parts of XML documents

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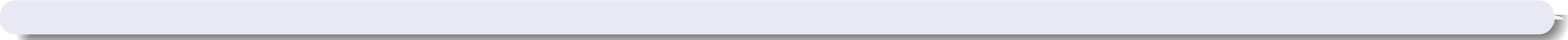
XML processing instruction

Parts of XML documents

⟨?xml version="1.0" encoding="ISO-8859-1" ?⟩

⟨?xml-stylesheet href="headlines.css" type="text/css" ?⟩

⟨invoice⟩
 ⟨product⟩Lenovo X201⟨/product⟩
 ⟨customer⟩Jack Ryan⟨/customer⟩
 ⟨amount⟩NZ\$1822⟨/amount⟩
 ⟨data⟩30-03-2011⟨/date⟩
⟨/invoice⟩



Parts of XML documents

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Definition of the Element root element

Name two ways in which Document Type Definitions (DTDs) can *assist XML parsers*.

How XML parsers use DTDs

- DTDs can ensure that all required XML elements are present in the XML document
- DTDs can prevent undefined XML elements from being used in the XML document
- DTDs enforce a specific data structure on the XML document
- DTDs specify the use of attributes of XML elements and define their permitted values

DTD Modeling

Consider the following XML document.

```
<Movies>
  <Movie Year-released="2009">
    <Title>The girl with the dragon tattoo</Title>
    <Director>Niels Arden Oplev</Director>
    <Actors>
      <Actor>Noomi Rapace</Actor>
    </Actors>
  </Movie>
  <Movie Year-released="2009">
    <Title>The girl who played with fire</Title>
    <Writer>Jonas Frykberg</Writer>
    <Writer>Stieg Larsson</Writer>
    <Actors>
      <Actor>Noomi Rapace</Actor>
      <Actor>Michael Nyqvist</Actor>
    </Actors>
  </Movie>
</Movies>
```

- 1 Write down a *Document Type Definition* (DTD) which validates the XML document.
- 2 Considering that DTDs are meant to model typical XML documents associated with the given domain, what changes would you make to your original DTD to validate typical XML documents that occur the movie domain. Argue why you make certain choices.

DTD Modeling - Validation only

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  <!ELEMENT Movies (Movie,Movie)>
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  <!ELEMENT Movie (Title,Director?,Writer?,
                  Writer?,Actors)>
  <!ATTLIST Movie Year-released
                  CDATA #REQUIRED>
  <!ELEMENT Title (#PCDATA)>
  <!ELEMENT Director (#PCDATA)>
  <!ELEMENT Writer (#PCDATA)>
  <!ELEMENT Actors (Actor,Actor?)>
  <!ELEMENT Actor (#PCDATA)>
]>
```

DTD Modeling - Validation only

```
<Movies>
  <Movie Year-released="2009">
    <Title>The girl with the dragon tattoo</Title>
    <Director>Niels Arden Oplev</Director>
    <Actors>
      <Actor>Noomi Rapace</Actor>
    </Actors>
  </Movie>
  <Movie Year-released="2009">
    <Title>The girl who played with fire</Title>
    <Writer>Jonas Frykberg</Writer>
    <Writer>Stieg Larsson</Writer>
    <Actors>
      <Actor>Noomi Rapace</Actor>
      <Actor>Michael Nyqvist</Actor>
    </Actors>
  </Movie>
</Movies>
```

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```

Meaningful DTD

```
<Movies>
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    <Actors>
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    <Writer>Jonas Frykberg</Writer>
    <Writer>Stieg Larsson</Writer>
    <Actors>
      <Actor>Noomi Rapace</Actor>
      <Actor>Michael Nyqvist</Actor>
    </Actors>
  </Movie>
</Movies>
```

```
<!DOCTYPE Movies [
  <!ELEMENT Movies (Movie+)>
  <!ELEMENT Movie (Title,Director?,
                    Writer*,Actors)>
  <!ATTLIST Movie Year-released CDATA
                  #REQUIRED>
  <!ELEMENT Title (#PCDATA)>
  <!ELEMENT Director (#PCDATA)>
  <!ELEMENT Writer (#PCDATA)>
  <!ELEMENT Actors (Actor+)>
  <!ELEMENT Actor (#PCDATA)>
]>
```

- There are many movies, at least one.
- There is at most one director for each movie, and
- any number of writers, and
- there is at least one actor but there could be many.

Meaningful DTD

```
<Movies>
  <Movie Year-released="2009">
    <Title>The girl with the dragon tattoo</Title>
    <Director>Niels Arden Oplev</Director>
    <Actors>
      <Actor>Noomi Rapace</Actor>
    </Actors>
  </Movie>
  <Movie Year-released="2009">
    <Title>The girl who played with fire</Title>
    <Writer>Jonas Frykberg</Writer>
    <Writer>Stieg Larsson</Writer>
    <Actors>
      <Actor>Noomi Rapace</Actor>
      <Actor>Michael Nyqvist</Actor>
    </Actors>
  </Movie>
</Movies>
```

```
<!DOCTYPE Movies [
  <!ELEMENT Movies (Movie+)>
  <!ELEMENT Movie (Title,Director?,
                    Writer*,Actors)>
  <!ATTLIST Movie Year-released CDATA
                  #REQUIRED>
  <!ELEMENT Title (#PCDATA)>
  <!ELEMENT Director (#PCDATA)>
  <!ELEMENT Writer (#PCDATA)>
  <!ELEMENT Actors (Actor+)>
  <!ELEMENT Actor (#PCDATA)>
]>
```

- There are many movies, at least one.
- There is at most one director for each movie, and
- any number of writers, and
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Meaningful DTD

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      <Actor>Noomi Rapace</Actor>
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      <Actor>Noomi Rapace</Actor>
      <Actor>Michael Nyqvist</Actor>
    </Actors>
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  <!ELEMENT Title (#PCDATA)>
  <!ELEMENT Director (#PCDATA)>
  <!ELEMENT Writer (#PCDATA)>
  <!ELEMENT Actors (Actor+)>
  <!ELEMENT Actor (#PCDATA)>
]>
```

- There are many movies, at least one.
- **There is at most one director for each movie**, and
- any number of writers, and
- there is at least one actor but there could be many.

Meaningful DTD

```
<Movies>
  <Movie Year-released="2009">
    <Title>The girl with the dragon tattoo</Title>
    <Director>Niels Arden Oplev</Director>
    <Actors>
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    </Actors>
  </Movie>
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```

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<!DOCTYPE Movies [
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    #REQUIRED>
  <!ELEMENT Title (#PCDATA)>
  <!ELEMENT Director (#PCDATA)>
  <!ELEMENT Writer (#PCDATA)>
  <!ELEMENT Actors (Actor+)>
  <!ELEMENT Actor (#PCDATA)>
]>
```

- There are many movies, at least one.
- There is at most one director for each movie, and
- any number of writers, and
- there is at least one actor but there could be many.

Meaningful DTD

```
<Movies>
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  </Movie>
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  <!ELEMENT Title (#PCDATA)>
  <!ELEMENT Director (#PCDATA)>
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  <!ELEMENT Actors (Actor+)>
  <!ELEMENT Actor (#PCDATA)>
]>
```

- There are many movies, at least one.
- There is at most one director for each movie, and
- any number of writers, and
- there is at least one actor but there could be many.

<code><enrolment></code>	<code><!ELEMENT enrolment (student⁺, class)></code>
<code> <student studentID="s01" /></code>	<code><!ELEMENT student EMPTY></code>
<code> <class studentIDrefs="s01 s02"></code>	<code><!ATTLIST student studentID ID #REQUIRED></code>
<code><title>ELCM XML</title></code>	<code><!ELEMENT class (title)></code>
<code></class></code>	<code><!ATTLIST class studentIDrefs IDREFS #REQUIRED></code>
<code></enrolment></code>	<code><!ELEMENT title (#PCDATA)></code>

Explain whether the XML document on the left conforms to the DTD on the right.

DTD Validation

```
<enrolment>  
  <student studentID="s01"/>  
  <class studentIDrefs="s01 s02">  
    <title>ELCM XML</title>  
  </class>  
</enrolment>
```

```
<!ELEMENT enrolment (student+, class)>  
<!ELEMENT student EMPTY>  
<!ATTLIST student studentID ID #REQUIRED>  
<!ELEMENT class (title)>  
<!ATTLIST class studentIDrefs IDREFS #REQUIRED>  
<!ELEMENT title (#PCDATA)>
```

- The document does *not* conform to the DTD
- It references the studentID "s02" which is not the value of any attribute of type ID in the document
- This, however, is a condition necessary for conformance

DTD Validation

<code><enrolment></code>	<code><!ELEMENT enrolment (student⁺, class)></code>
<code> <student studentID="s01" /></code>	<code><!ELEMENT student EMPTY></code>
<code> <class studentIDrefs="s01 s02"></code>	<code><!ATTLIST student studentID ID #REQUIRED></code>
<code><title>ELCM XML</title></code>	<code><!ELEMENT class (title)></code>
<code></class></code>	<code><!ATTLIST class studentIDrefs IDREFS #REQUIRED></code>
<code></enrolment></code>	<code><!ELEMENT title (#PCDATA)></code>

- The document does *not* conform to the DTD
- It references the studentID "s02" which is not the value of any attribute of type ID in the document
- This, however, is a condition necessary for conformance

Write down a *DTD* that specifies an element with *mixed content*.

```
<?xml version="1.0" standalone="yes" ?>
```

```
<!DOCTYPE rootelement [  
  <!ELEMENT rootelement (#PCDATA|childelement)*>  
  <!ELEMENT childelement (#PCDATA)>]]>
```

```
<rootelement>
```

PCDATA Content

```
<childelement>Child element</childelement>
```

```
</rootelement>
```